

EVP PARAMETERS IDENTIFICATION PROCEDURE

To follow this procedure, you need to have access to your identified EPNL kinematic hardening parameters and you also need the “.mast” document used on Zset to identify them. Those files will be the starting point of this procedure to identify EVP model parameters.

I. PRESENTATION OF EVP MECHANICAL MODEL

EVP model compare to EPNL kinematic model has only two more parameters to identify: K and n , referring to the material's viscous behavior. There is two experimental ways to show the viscous effect:

- With traction tests at different imposed strain rates.

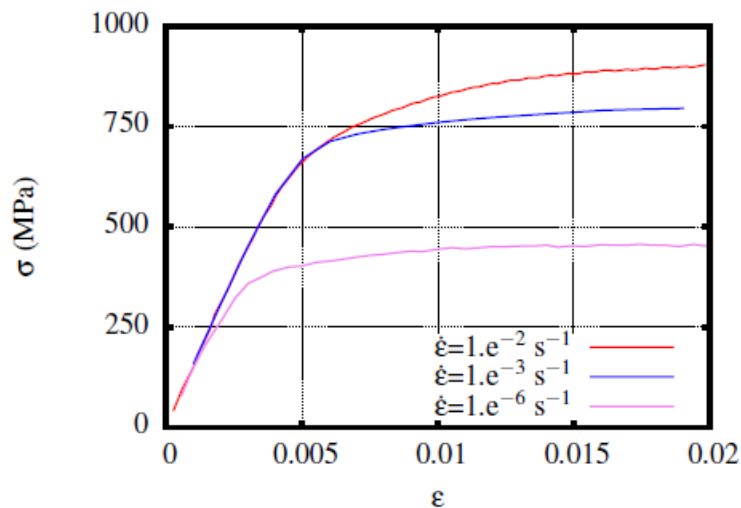
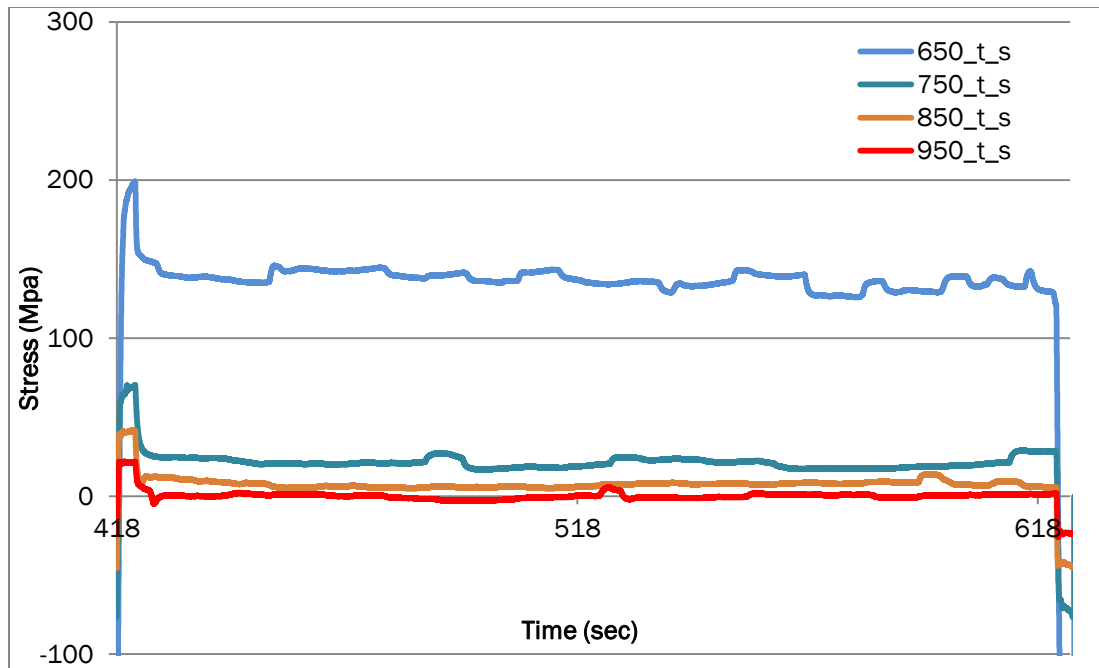


FIGURE 1 – Essais de traction à différentes vitesses de déformation imposée

- With a stress relaxation test within a significant dwell time (here 200sec) at different temperatures, called next trapezoidal signal.



Viscous effects on steel materials appears at temperature higher than 650 °C, this is why we have stress relaxation experimental data starting at this temperature.

In this procedure will be detailed different ways to identify viscous parameters using the second possibility, stress relaxation test.

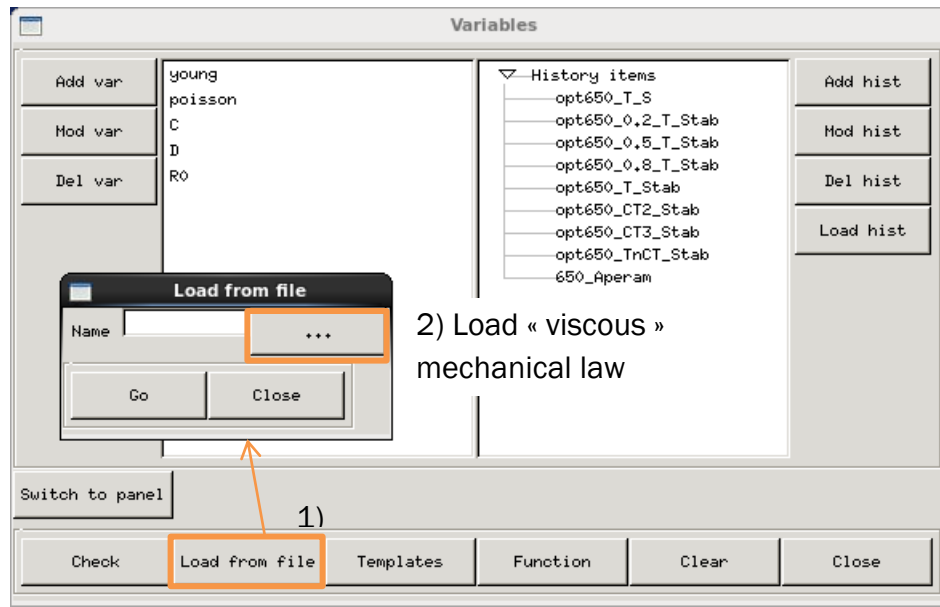
Procedures detailed in this document can be improved as there are some issues. After the detail of each procedure, the main issues and some ideas of improvements will indicated. Possible improvements are sometime complicated to take into account in the procedure, they have to make the procedure easier to use and more important to give correct identified parameters.

First thing to do here, as said earlier, is to open the EPNL kinematic hardening identification file corresponding to the considered temperature (from 650 °C to 950 °C).

From here, we want to transform this EPNL file into an EVP file.

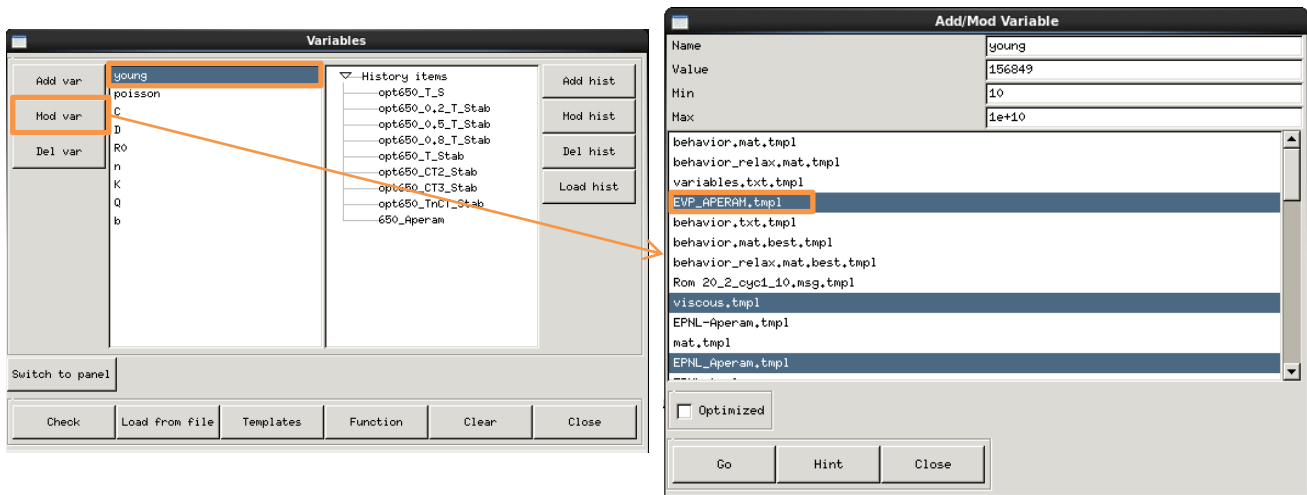
VARIABLES TOOL

To do so, we will first add K and n, viscous parameters, to the existing model. In the “Variables” tool, you need to charge a new mechanical model using mechanical law file call “viscous”.



There are now 9 parameters in your variables window: young, poisson, C, D, R0, K, n, Q and b. Q and b will not change during the identification as we do not consider isotropic hardening.

²Next step after charging the new parameters is to link them with the EVP_Aperam model. To do so you need for each parameter (except for Q and b not include in this model) to follow the next steps:

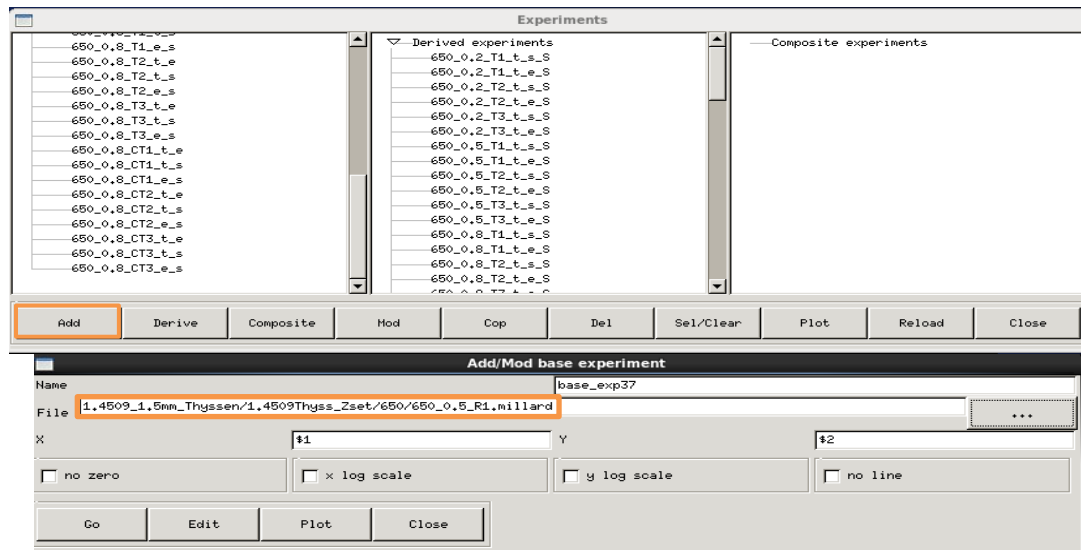


- 1) Select the parameter to modify
- 2) Click on "Mod var"
- 3) Select EVP_Aperam model without unselect the others models

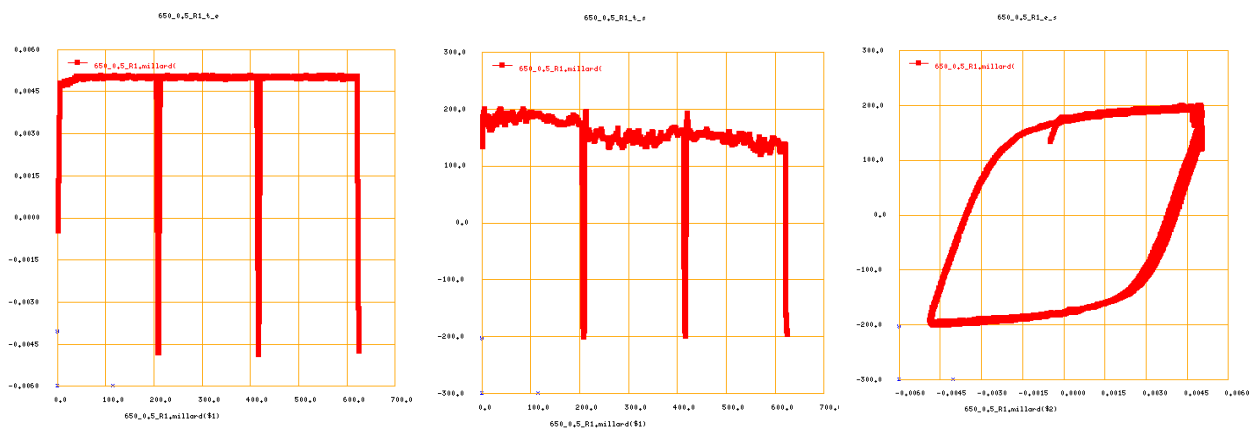
Do not forget to save your .mast file under a new name if you want to keep both files separately.

EXPERIMENT TOOL

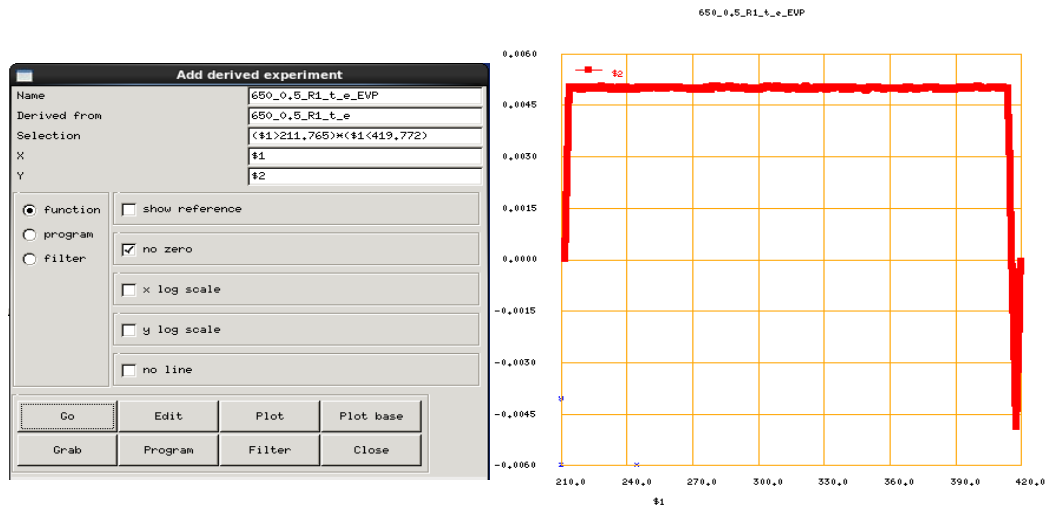
As explain in the presentation, we need other experimental data to identify the new parameters K and n. Those experiments are available in the usual directory and you can find them easily as they got an R in their name. Adding new experiments can be made like before.



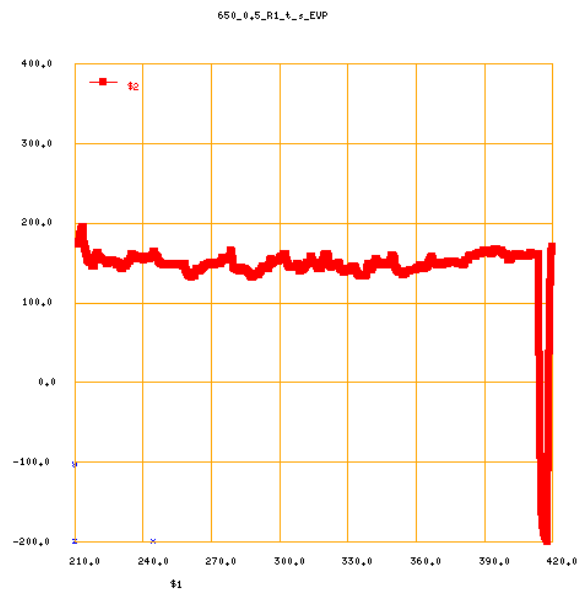
Select the experiment you want to study and you can plot the same graphics as before: time-eto, time-sig and eto-sig. These are the kind of curve we should have:



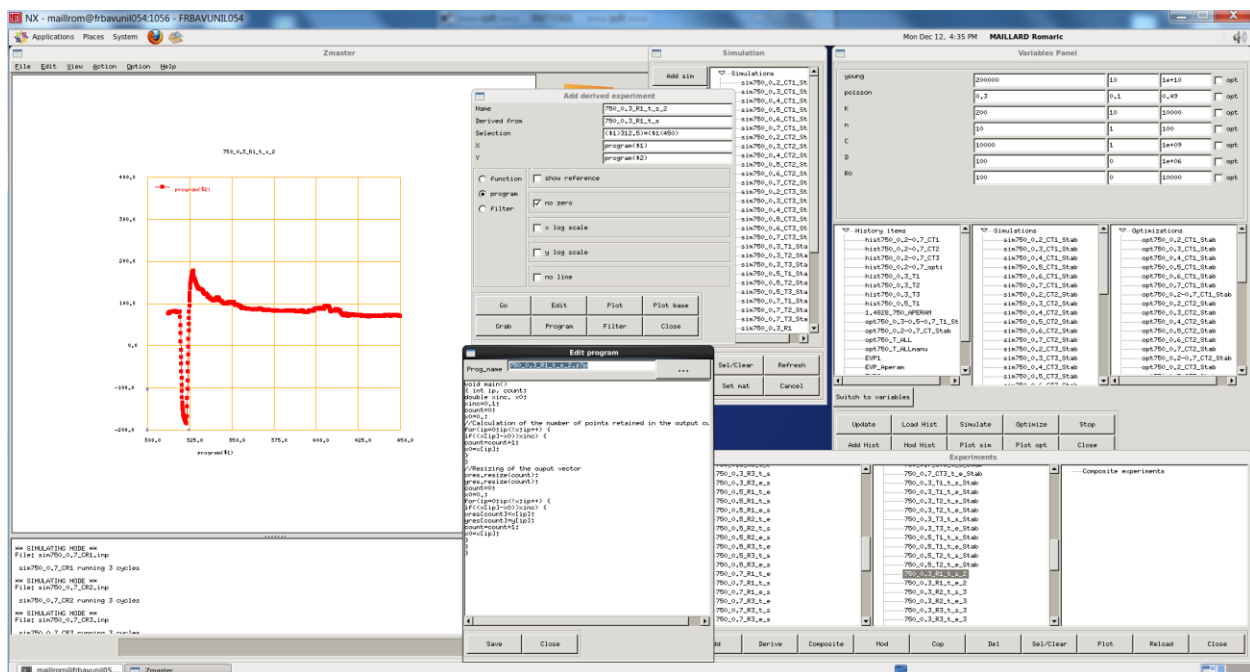
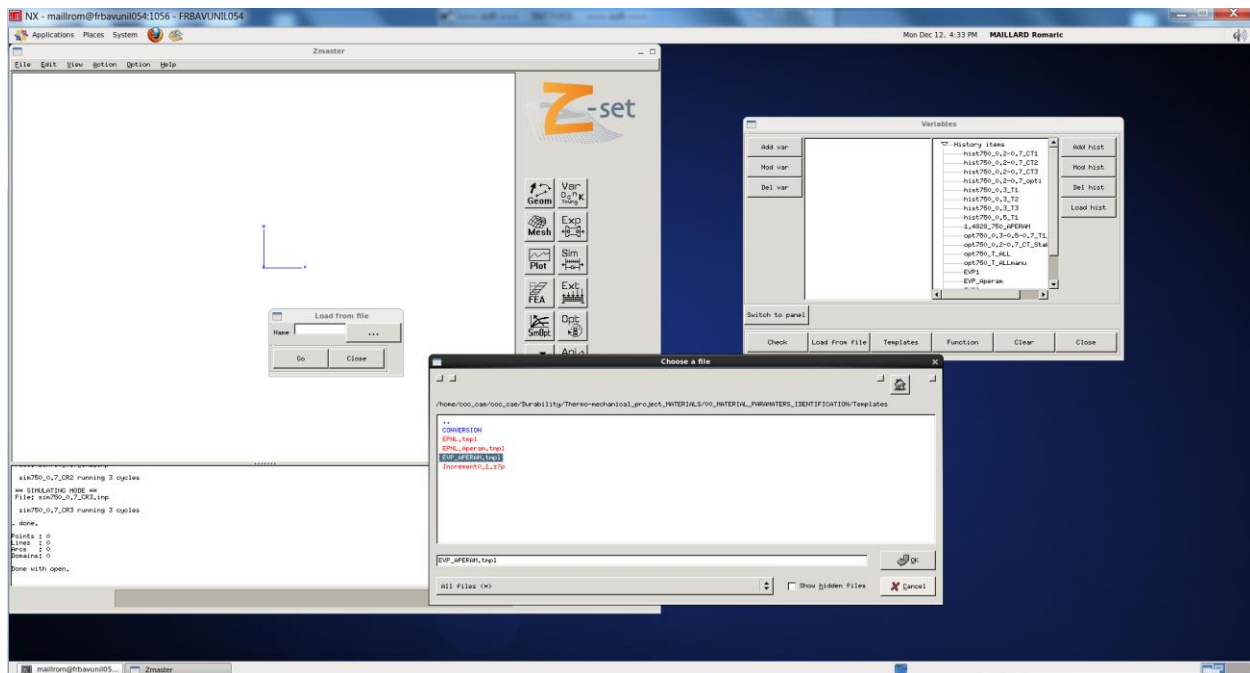
As this signal has multiple points, we need to derive this initial signal in order to take the best portion of signal to repeat. In this example, we will consider only the second part of signal. We select part of the signal as follow, starting with the t-e curve:

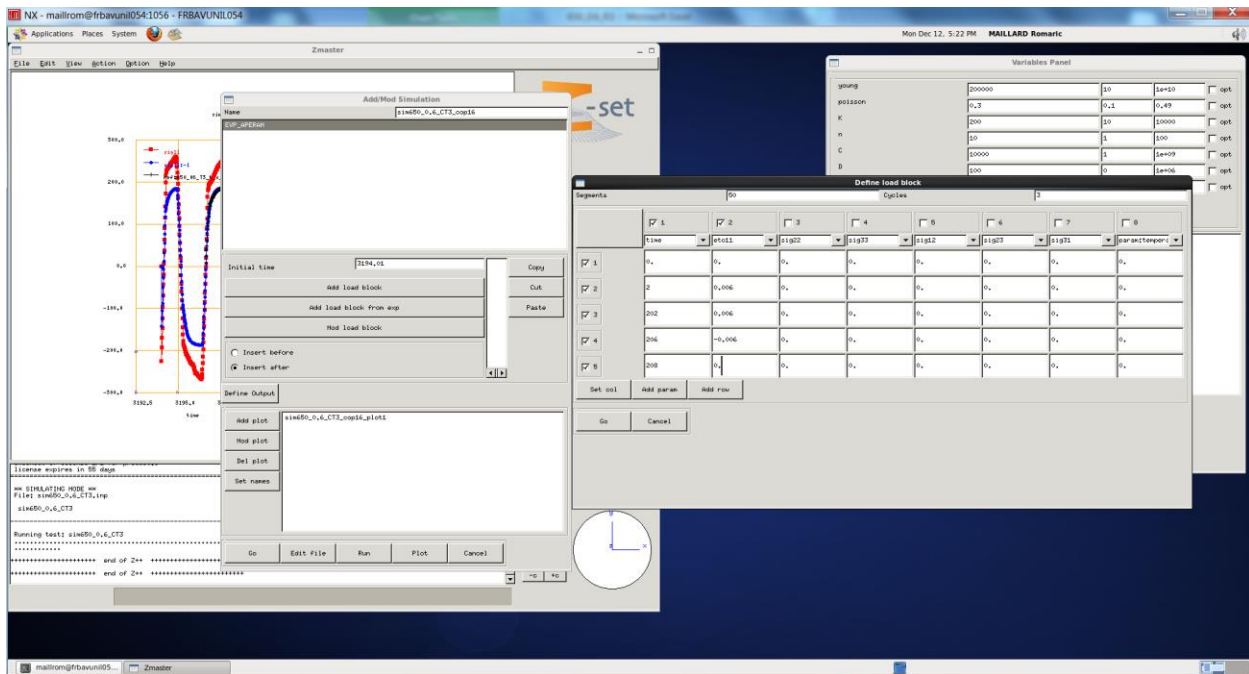
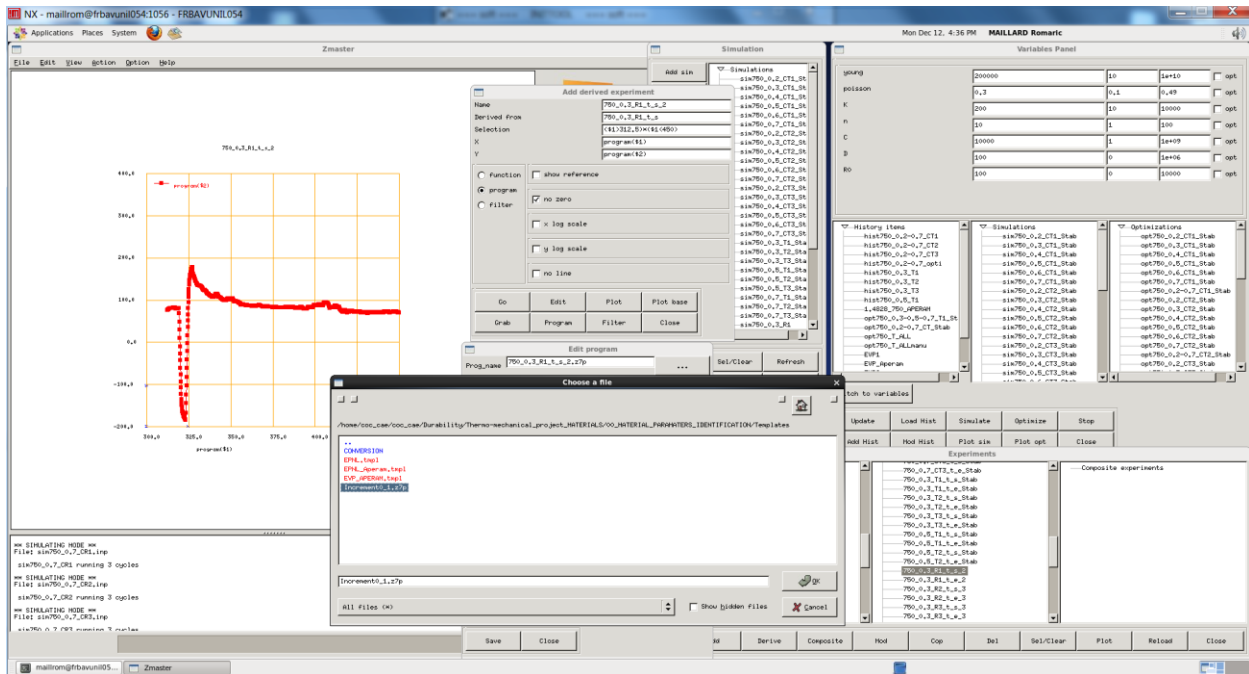


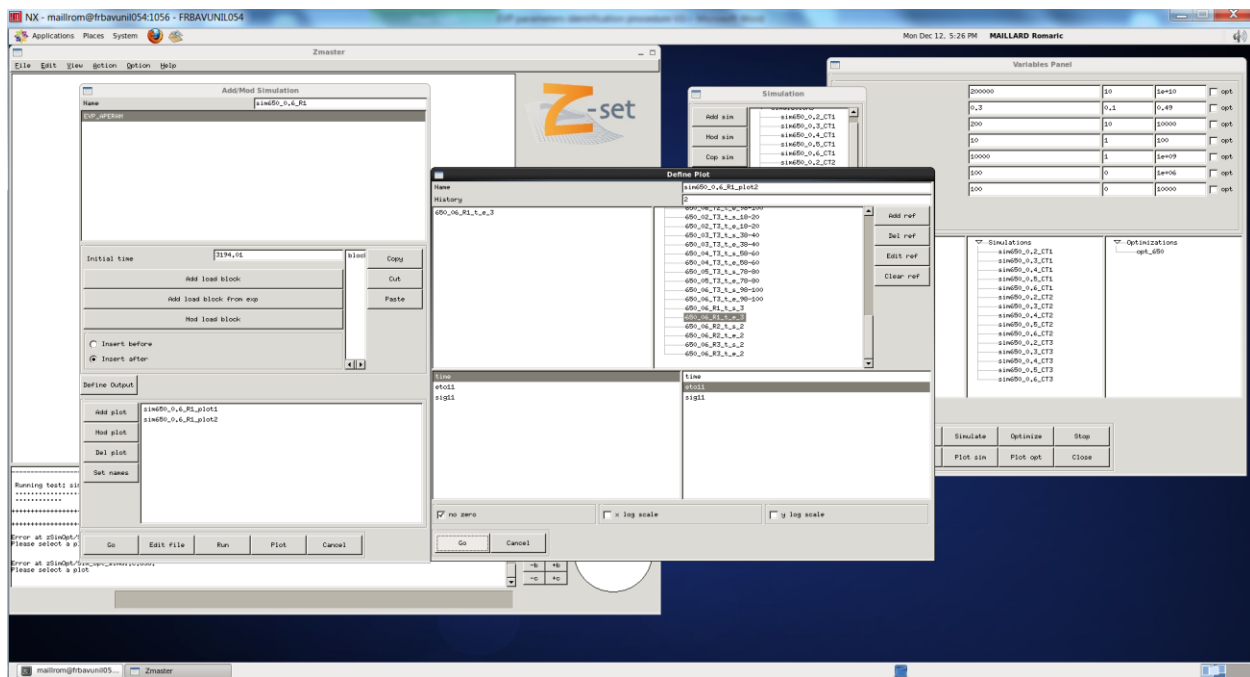
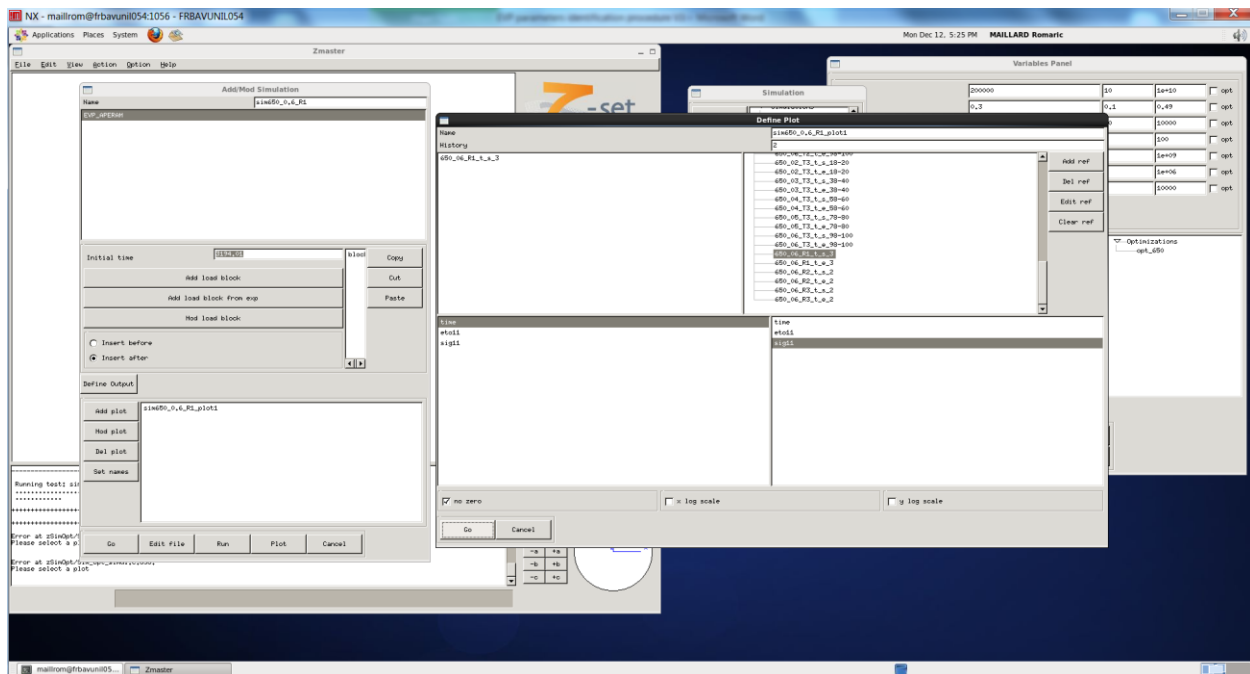
Open Add derived experiment window and select the part of signal selected as shown on previous graphic and complete the selection space with the first and the last point coordinate. Then you just have to copy this derived experiment and to change t-e curve into t-s curve. This is what we have then:

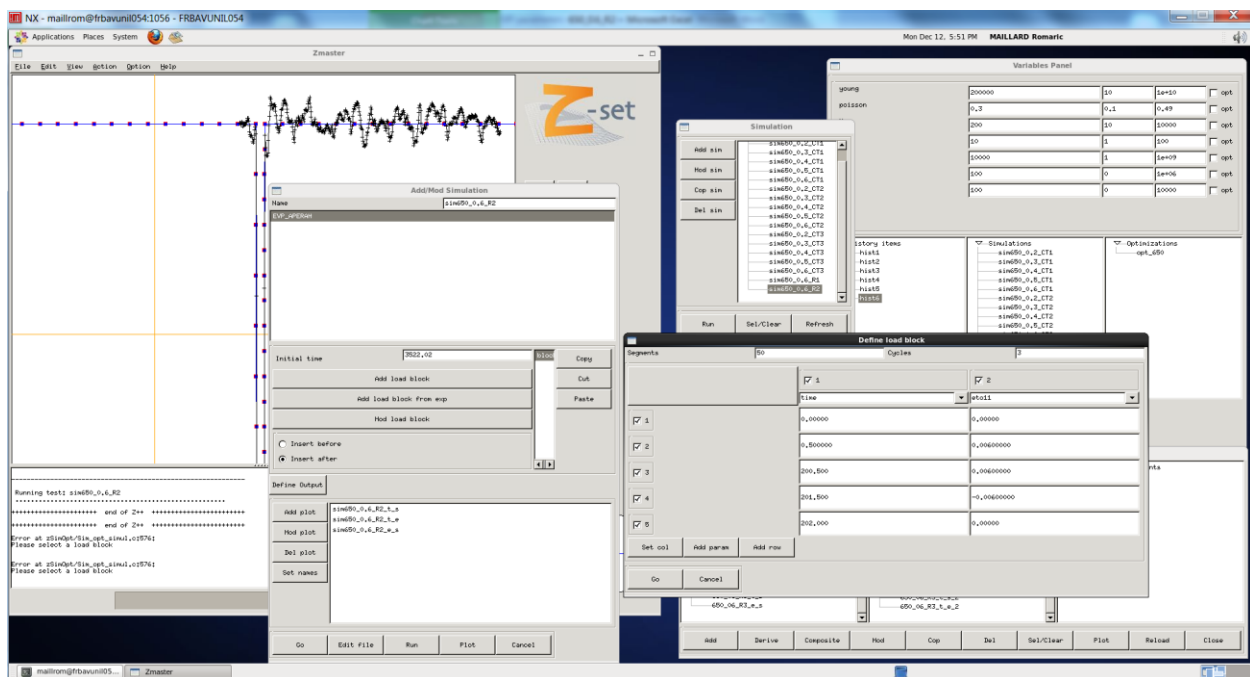
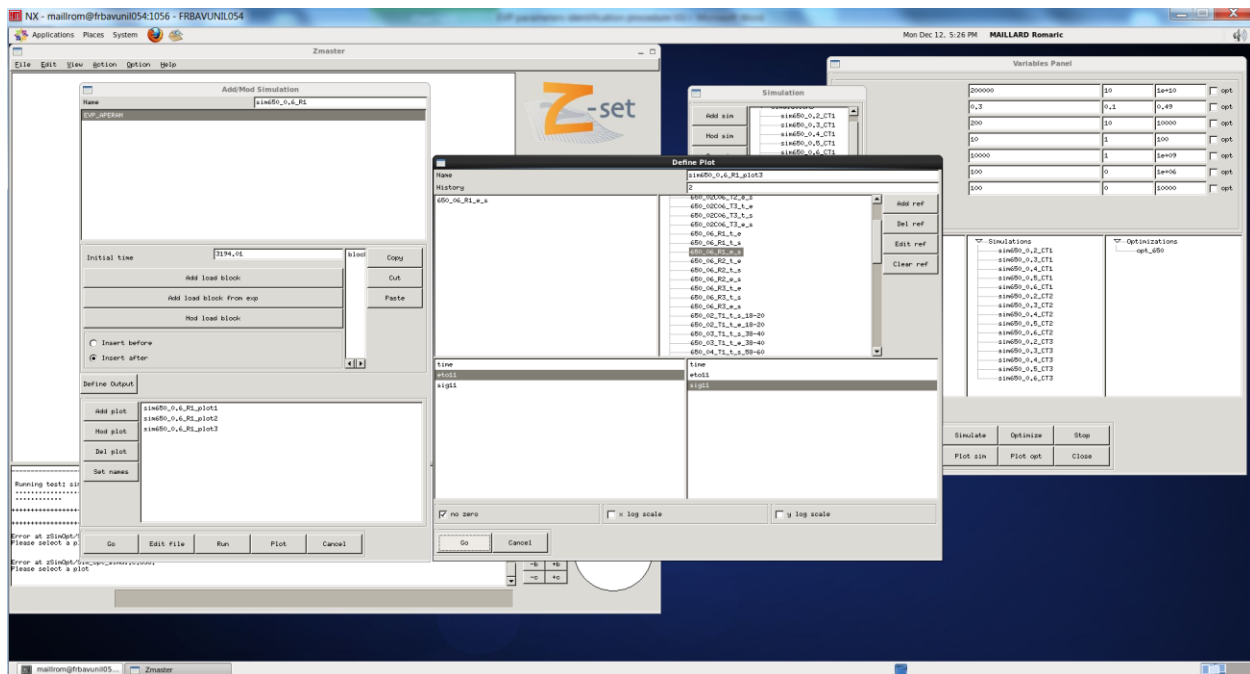


PROGRAM









II. VISCOUS PARAMETERS IDENTIFICATION PROCEDURE 1

First procedure possible to identify EVP parameters is the more complete one. This procedure will take into account both experimental signals, triangular and trapezoidal.

SIMULATION TOOL

Next thing to do is to create two simulations: 1 for the triangular signal and 1 for the trapezoidal signal. The first simulation, triangular signal, is defined as before:

Segments	1	2	3	4	5	6	7	8
time	0.	0.	0.	0.	0.	0.	0.	0.
etot1	0,597	0,005	0.	0.	0.	0.	0.	0.
sig22	1,841	-0,005	0.	0.	0.	0.	0.	0.
sig33	2,489	0.	0.	0.	0.	0.	0.	0.
sig12	0.	0.	0.	0.	0.	0.	0.	0.
sig23	0.	0.	0.	0.	0.	0.	0.	0.
sig31	0.	0.	0.	0.	0.	0.	0.	0.
param:temper	0.	0.	0.	0.	0.	0.	0.	0.

The right screenshot also shows a list of segments with checkboxes for each, and a 'Cycles' field set to 100. The 'Define Output' section at the bottom of the right dialog is empty.

- 1) Type a file name
- 2) Choose EVP_Aperam simulation template
- 3) Add the initial time from the Stab t-e experiment
- 4) Click Add block
- 5) Choose 5 segments to have complete hysteresis
- 6) Choose 100 cycles to simulate a complete signal
- 7) Complete these two columns using significant points from the experiment in order to reproduce the real signal

The first signal is now define and we need now to create another simulation using the same method for the trapezoidal signal. This second simulation needs to be completed with the derived experiment as next:

Define load block

Segments: 1) 5 Cycles: 2) 3

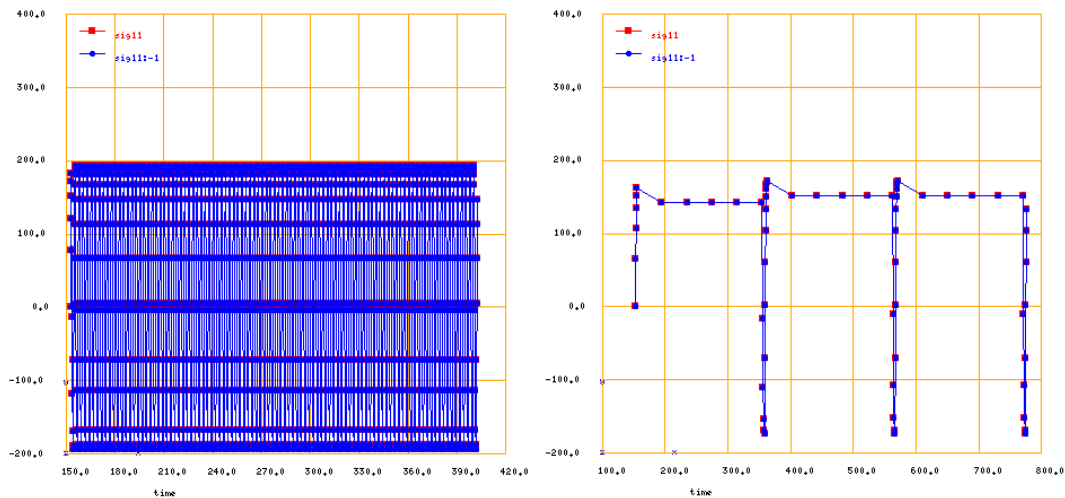
	1	2	3	4	5
time	eto11	sig22	sig33	sig12	
1	0,00000	0,00000	0,00000	0,00000	0,00000
2	2,14800	0,00500000	0,00000	0,00000	0,00000
3	201,836	0,00500000	0,00000	0,00000	0,00000
4	205,996	-0,00500000	0,00000	0,00000	0,00000
5	207,998	0,00000	0,00000	0,00000	0,00000

Set col Add param Add row

Go Cancel

- 1) Type 5 in the segments blank space to have a good amount of points to plot the complete response
- 2) Choose to repeat the load 3 times in order to reproduce the complete signal
- 3) Add the values of time and strain in the table using the significant points coming from the derived experiment

Doing that gives you two simulations with two different signals and these are the shape of the time-sig plots given with these simulations.



OPTIMIZATION TOOL

Now that both simulations are created, we need to create the optimization corresponding. In the Optimization tool you need to create 1 optimization using both simulations. This is how you can do that:

Add/Mod Optimization

Name: opt650_0,5_EVP

levenberg_marquardt

Add comp (1)

Mod comp

Del comp

Go Edit inp Plot comp Cancel

Add/Mod Comparison

Height: 1,0

☒ active

t_file_file

sim650_0,2_CT3_Stab

sim650_0,3_CT3_Stab

sim650_0,4_CT3_Stab

sim650_0,5_CT3_Stab

sim650_0,6_CT3_Stab

sim650_0,2_T1_Stab

sim650_0,2_CT1_Stab

sim650_0,4_CT1_Stab

sim650_0,5_CT1_Stab

sim650_0,6_CT1_Stab

sim650_0,2_CT1_S

sim650_0,2_CT2_S

sim650_0,2_CT3_S

sim650_EVP_T (2)

sim650_EVP_R

Experiments

650_0,2_T1_t_e

650_0,2_T1_t_s

650_0,2_T2_t_e

650_0,2_T2_t_s

650_0,2_T3_t_e

650_0,2_T3_t_s

650_0,5_T1_t_e

650_0,5_T1_t_s (3)

650_0,5_T1_e_s

650_0,5_T2_t_e

650_0,5_T2_t_s

time

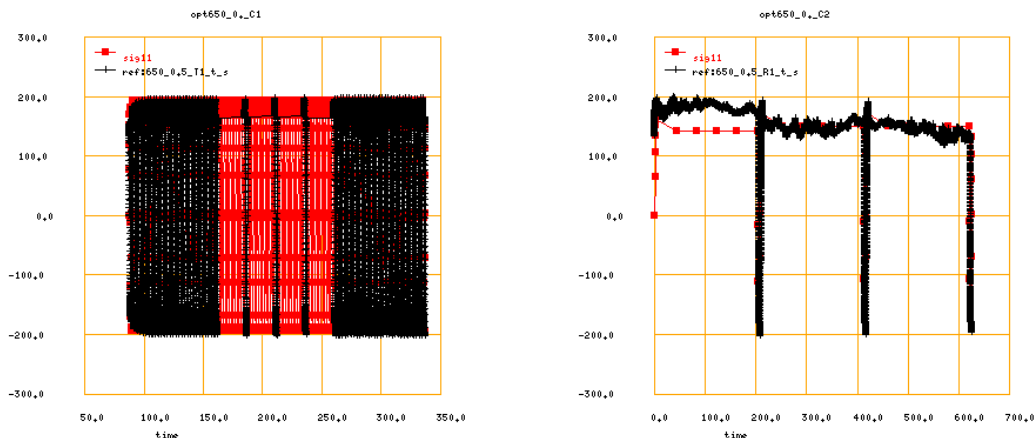
sig11 (4)

Go (5) Set weights Cancel

- 1) Create a new optimization and click on Add comp
- 2) Select the first simulation created
- 3) Select the corresponding experiment
- 4) Select time and sigma
- 5) Click Go

You need to follow the same steps for the second component using the second simulation created.

These are the plot you should have:



We notice that both plot experimental and simulation are superimposed well. If this is not the case, you need to check the initial time written in the simulation block.

Now, everything is ready for the first procedure. You need to load the average EPNL parameters identified before and to launch the optimization on 4 parameters at the same time: K, n, C and D.

Main issue with this procedure is the time calculation. As we use both complete signals, there are too many points to analyze and the optimization is taking too much time (several hours). This procedure gives good parameters and good correlation with experimental data.

In order to reduce the time calculation of this procedure, these are the main ways possible:

- To use only the last 3 cycles of the triangular signal only 1 cycle of the trapezoidal signal.
- Filter the experimental data to reduce the number of points to consider during the optimization.
- To redo the experimental test in order to have a signal with less noise, as for the LD tests.
- To optimize parameters differently

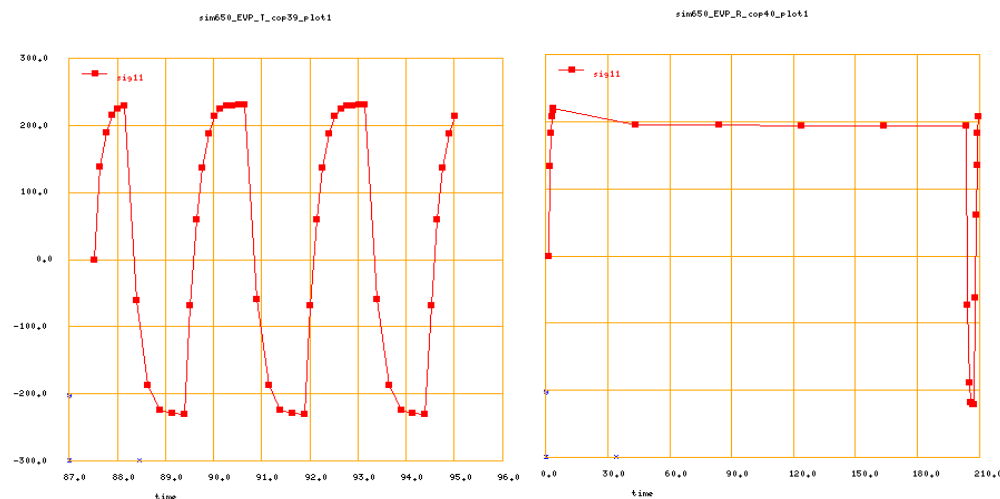
III. VISCOUS PARAMETERS IDENTIFICATION PROCEDURE 2

This second procedure is based on the conclusion made with the first procedure. We will try to apply the first possible improvement and to see what the conclusion is.

In this procedure we will consider only the last 3 cycles for the triangular signal and the second cycle for the trapezoidal signal.

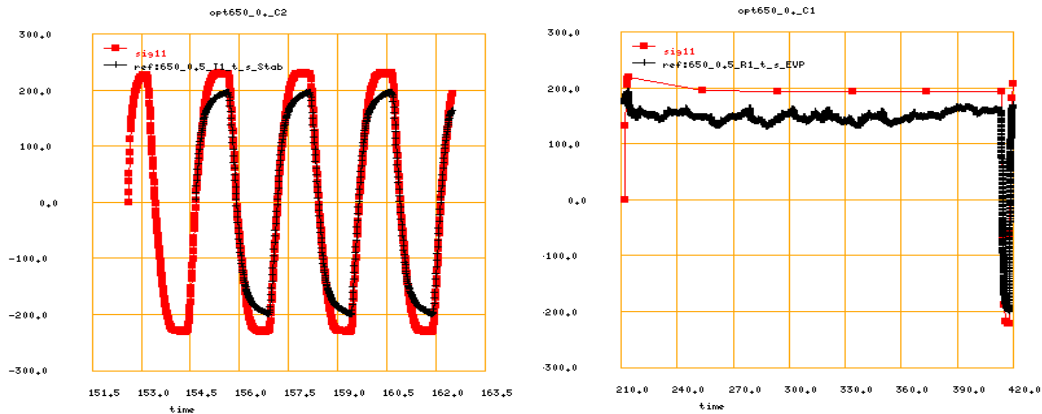
SIMULATION TOOL

The simulations for this procedure will be created following the same steps as before. The only things that changed are the number of cycles we consider. These are the time-sigma curves you should have from the simulations:



OPTIMIZATION TOOL

Now that you created the simulation for this procedure, you can use them to create the optimization. In the optimization tool you need to select the experiment curve corresponding to your simulation. This will give you the next curves:



Then there are many possible way to use this optimization to identify correct parameters:

- 1) As in the first procedure, we can optimize every parameter at once (K , n , C and D). Using this does not give us good results as we do not play on the n value.
- 2) We can keep n constant at first at a theoretical value, optimize K and R_0 first, next C and D and finally all together (C , D , R_0 , K and n). This possibility gives us correct results but take some time to calculate.

Comparing to the first procedure, the time calculation is well reduced (from several hours to a few minutes). The main issue here is to define the combination of parameters to identify.